

# Logarithmic Differentiation

section 5.1

11/12/25

$$y = \frac{x^3 \sqrt{5x^2 - 4}}{(x^2 + 1)^{3/2}}$$

$$\ln y = \ln \left( \frac{x^3 \sqrt{5x^2 - 4}}{(x^2 + 1)^{3/2}} \right)$$

$$\ln y = 3 \ln(x) + \frac{1}{2} \ln(5x^2 - 4) - \frac{3}{2} \ln(x^2 + 1)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{3}{x} + \frac{5 \cancel{10} x}{2(5x^2 - 4)} - \frac{3(\cancel{2}x)}{2(x^2 + 1)}$$

$$\frac{dy}{dx} = \left( \frac{3}{x} + \frac{5x}{5x^2 - 4} + \frac{3x}{x^2 + 1} \right) y$$

$$\boxed{\frac{dy}{dx} = \left( \frac{3}{x} + \frac{5x}{5x^2 - 4} + \frac{3x}{x^2 + 1} \right) \left( \frac{x^3 \sqrt{5x^2 - 4}}{(x^2 + 1)^{3/2}} \right)}$$

$$y = 2^x$$

$$\ln y = \ln(2^x)$$

$$\ln y = x \ln(2)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \ln(2)$$

$$\frac{dy}{dx} = \ln(2) \cdot y$$

$$\boxed{\frac{dy}{dx} = \ln(2) \cdot 2^x}$$

$$y = x^3$$



Power rule!

$$y = 3^x$$



exponential rule!

$$y = x^x$$



logarithmic differentiation

$$y = x^{2x}$$

$$\ln y = \ln(x^{2x})$$

$$\ln y = 2x \ln(x)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = 2 \cdot \ln x + \frac{1}{x} \cdot 2x$$

$$\frac{dy}{dx} = (2 \ln x + 2) x^{2x}$$

$$y = x^{\tan x}$$

$$\ln y = \ln(x^{\tan x})$$

$$\ln y = \tan x \cdot \ln x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \sec^2 x \cdot \ln x + \frac{\tan x}{x}$$

$$\frac{dy}{dx} = \left( \sec^2 x \ln x + \frac{\tan x}{x} \right) x^{\tan x}$$